OPERATION AND MAINTENANCE MANUAL

PROJECT: AD 9039 – The Dubai Mall Za'abeel Expansion (TDMZE)	
Date: 3 February 2019	**TDMZE-AFG-AR-MAN-00-AL-0089**
To: Mr. Abdul Rehman Qureshi –Project Manager - BSBG Cc: Mr. Tom Dundas – Project Manager's Representative - Turner	From: Contractor Representative Dutco Construction Co.LLC

ENCLOSED DOCUMENTS

Description	Rev.	Issue Status	No. of copies	Document Reference
Draft Operation and Maintenance Manual for Scissor Lift	0	4	1 sets	----------

Issue status:	1 - for record	2 – for review and comment	3 – for action	4– for approval

Contractor Representative:

Signature:

ACKNOWLEDGEMENT BY RECIPIENT (completed form to be returned to the DBB document controller)

"I confirm receipt of the above listed documents"

Comments:

☐ Approved ☐ Approved As Noted ☐ Revise & Resubmit ☐ For Information

BSBG Comments
1. Refer marked-up comments.
2. The manual should include the power pack and control panel as its part of the system.

BSBG Brewer Smith Brewer Gulf

	A - No Comment
	B - Comments as Noted
✓	C - Unacceptable, Re-Submit

The Architect/ Engineer's review of this material/ drawing/ document is for the purpose of ascertaining conformity with the basic concept, profiles and general arrangements only. The review shall not be construed to mean that the Architect/ Engineer accepts final installation detail, responsibility for which will remain with the Contractor. The Contractor is responsible for any errors or ommissions in the material sample/ drawing/ document and for meeting the requirements of the Contract documents.

Qureshi
06.03.2019

Name: Location:

Signature: Date:

DBB mailing address: **Dutco Construction Co. LLC, PO Box 8944, Dubai, UAE**



TDMZE

THE DUBAI MALL ZA'ABEEL EXTENSION

OPERATIONS & MAINTENANCE MANUAL

VOLUME 1 of 1

PART 1 of 1

SCISSOR LIFT

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Include page number on each item.

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Operating manual and literature

SECTION 1

1. ABOUT THE PROJECT.....2

(Typical Comment) Page numbering is incorrect. Fix the page numbering all throughout the manual.

1. ABOUT THE PROJECT

The Dubai Mall Za'abeel Extension (TDMZE) is an expansion to the existing Dubai Mall which increases the footprint of the mall by 40% by connecting the two structures with bridges that cross the Financial Centre Road elevated multilane highways at a high level (level 6).

TDMZE consists of retail and car parking.

TDMZE building provides additional parking to the existing Dubai Mall as well as a substantial increase in retail (including 3 new anchor tenants), food and beverage areas and restaurants. There is also provision for a future 50+ storey residential tower to be integrated into this project.

The floor-wise space breakdown is provided below:

2. P1: Ground Floor

- Utility Rooms, Fire Command Center, Fire Corridors, Police Rooms and MEP Services Rooms and Ramp Access
- Retail along South Elevation
- Loading Docks
- Service Lifts, Main Lobby, Toilets and Lift Lobbies.
- DEWA Corridor

3. P2: Mezzanine

- Typical Carpark Floor
- Lift Lobbies and Stairs
- MEP Service Rooms

4. P3: First Floor

- Typical Carpark Floor
- Public and Residential Parking
- Escalator Lobbies and Taxi Waiting Area
- Lift Lobbies and Stairs
- MEP shafts and Riser

5. P4: Second Floor

- Typical Carpark Floor
- Public and Residential Parking To be fixed
- Escalator and Lift Lobbies
- Male and Female Prayer rooms, Ablution and Public Toilets
- Stairs and MEP Service Rooms

6. P5: Third Floor

- ∉ Retail and Carpark Floor
- ∉ Escalator and Lift Lobbies
- ∉ Stairs and MEP Service Rooms

7. P6: Fourth Floor

- ∉ Retail and Carpark Floor
- ∉ Flagship Tenant, F&B, Foodcourt, Entertainment and Public Toilets
- ∉ Ped Link Bridge to The Dubai Mall (NIC)

8. P7 & P8 Fifth and Sixth Floors

- ∉ Retail and Carpark Floor
- ∉ Escalator and Lift Lobbies
- ∉ Stairs and MEP Service Rooms

9. P9: Roof Slab

- ∉ Typical Carpark Floor
- ∉ Escalator and Lift Lobbies
- ∉ Stairs and MEP Service Rooms
- ∉ Residential Amenities (NIC)

SECTION 2

2. CHANGE CONTROL 2

2. CHANGE CONTROL

DATE	SYSTEM	MANUAL SECTION	DESIGN BY	DETAILS OF CHANGE
Jan 2019	Mechanical	SCISSOR LIFT	NIHVA TECHNOLOGIES PVT. LTD.	Revision 00

Date should be complete (**Day**/Month/Year)

SECTION 3

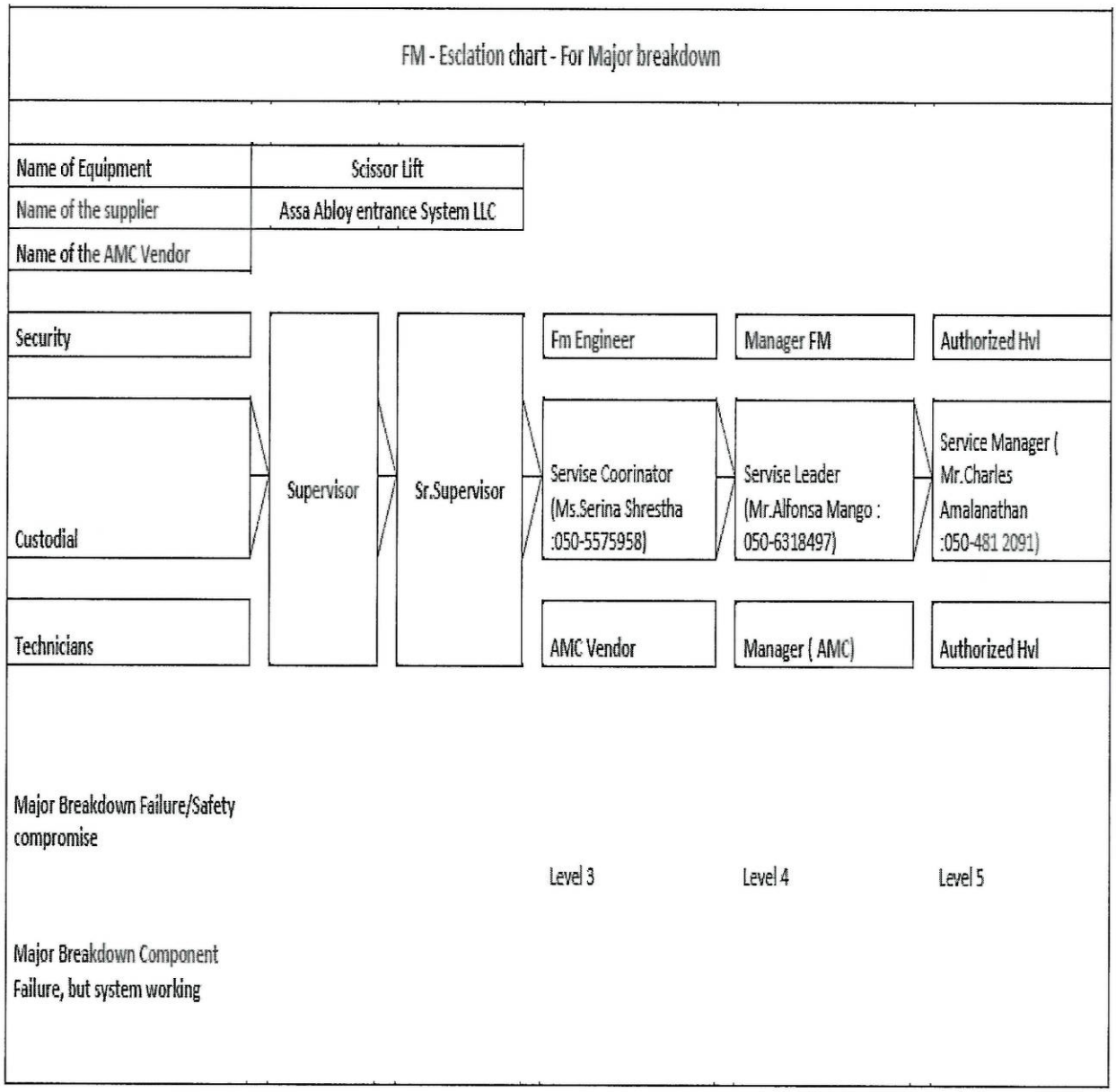
3. EMERGENCY ESCALATION PROCEDURE 3

3.1 FM ESCALATION CHART FOR MAJOR BREAKDOWNS 3.1

3.2 EMERGENCY CONTACTS 3.2

Not available on this section

3.1 FM Escalation Chart for Major Breakdowns



SECTION 4

4. EMERGENCY INFORMATION 2

4.1 EMERGENCY CONTACTS 2

4.2 PROJECT DIRECTORY 2

4. EMERGENCY INFORMATION

4.1 Emergency Contacts

In case of	Contact	Telephone
Accident or Incident	Police	Tel: 999
Fire	Fire Brigade	Tel: 997
Medical Emergency	Ambulance Or Police	Tel: 998 / 999
Power Failure	DEWA	Tel: 04-3244 444 Fax: 04-3248 111 Email: csd@dewa.gov.ae
Telephone Enquiry	ETISALAT	Tel: 181

4.2 Project Directory

Role	Organisation Name & Address	Phone/Fax & E-Mail Details
Employer	Emaar Properties PJSC P O Box 9440 Dubai United Arab Emirates	Tel: +971 4 367 3333 Fax: +971 4 367 3000 E-mail: ffadda@emaar.ae
Local Authority	Dubai Municipality P O Box 67 Dubai United Arab Emirates	Tel: +971 4 221 5555 Fax: +971 4 224 6666 E-mail: info@dm.gov.ae
Project Manager	Turner Construction - International Middle East Ltd. P.O. Box 13036 Dubai, United Arab Emirates	Tel: +971 4 3675609 Fax: +971 4 3348216 E-mail:
Lead Consultant	Brewer Smith Brewer Gulf PO Box 50667 Dubai United Arab Emirates	Tel: +971 4 294 2042 Fax: +971 4 294 2101 E-mail: enquiries@bsbgulf.com

Role	Organisation Name & Address	Phone/Fax & E-Mail Details
Quantity Surveyor	Omnium International Limited P.O. Box 62256 Dubai, United Arab Emirates	Telephone: +9714 3434600 Fax: +971 4 3437142 Email: dubai@omniumint.com
Main Contractor	Dutco Construction Co LLC P O Box 8944 Dubai United Arab Emirates	Tel: +971 4 880 1010 Fax: +971 4 880 1515 E-mail: mail@dbb.ae
Installing Contractors Details	ASSA ABLOY Entrance Systems P.O.Box 80983 Dubai United Arab Emirates	Tel: +971 4 8852888 Fax: +971 4 8854919 E-mail: info.ae.entrance@assaabloy.com
Maintenance Contractors Details	Will be decided at a later stage	

1. SECTION 5

5. EMERGENCY EQUIPMENT LOCATIONS 2

5. EMERGENCY EQUIPMENT LOCATIONS

For the location of general emergency equipment such as **fire alarm call points, etc**. Please refer to the **Fire Detection and Alarm System O&M manual** and the **Fire Fighting System O&M manual**

SECTION 6

6. SYSTEM PURPOSE AND SCOPE 2

6.1 PURPOSE OF THE INSTALLATION 2

6. SYSTEM PURPOSE AND SCOPE

6.1 Purpose of the Installation

The NIHVA hydraulic scissor lift ensures below purpose.

1. Provide safe lifting to the loading bay area.
2. Provides most economic dependable & versatile methods of **lifting** loads...
3. This lift table raises load smoothly to any desired height.

SECTION 7

SECTION 7 1

7. CONTRACTUAL & LEGAL INFORMATION 2

7.1 CONTRACTUAL & LEGAL INFORMATION 2

7.2 TEST CERTIFICATES 2

7.3 OPERATIONAL HAZARDS 2

7.4 HAZARD PROTECTION 2

7.5 SAFETY PRECAUTIONS 2

No details provided for this section.

7. CONTRACTUAL & LEGAL INFORMATION

7.1 Contractual & Legal Information

Legal information	
System Ownership	EMAAR Properties PSJC
System Address	The Dubai Mall Zaabeel Expansion
Design Team	Brewer Smith Brewer Gulf (BSBG)
Consultants	~~TURNER~~ & BSBG
Installation Contractors	ASSA ABLOY Entrance Systems
Sub-Contractor(s)	ASSA ABLOY Entrance Systems
Installation Start Date	Installed
Practical completion date	Finished
DLP Expiry Date	From the date of the taking over certificate TBC
Warranty Expiry Date	The date of the taking over certificate
Warranty / DLP Agent	ASSA ABLOY Entrance Systems

7.2 Test Certificates

Please refer to attached certificates. Certificates to be attached within the related section.

7.3 Operational Hazards

- Lift should not be loaded beyond its rated load capacity.
- Don't lose the oil hoses without resting the lift on its maintenance stoppers.
- Do not operate damaged lift.
- Don't stand at the edges of lift platform while the lift is lowering.
- Don't change lowering valve setting without supervision or any assistance.
- In case of power failure, to resume the downward operation after Power ON, do not lower the lift. Instead, take the lift slightly upward & then resume to lower it.
- Don't put any manual stopper or anything in the roller track.

7.4 Safety Precautions

Use trained personnel for loading & unloading of equipment. The equipment must be properly handled when transferring from the unloading area to final site to prevent damage.

DO NOT operate equipment unless it is properly anchored. It may cause serious injury or death.

a) Install all safety devices, guards and guarding prior to equipment start-up.

b) Only Trained & Authorized person must use the lift.

c) Always use mechanical stopper if you are going to work in elevated condition.

d) Keep working area free from oil to Avoid Slipping/Fire hazards.

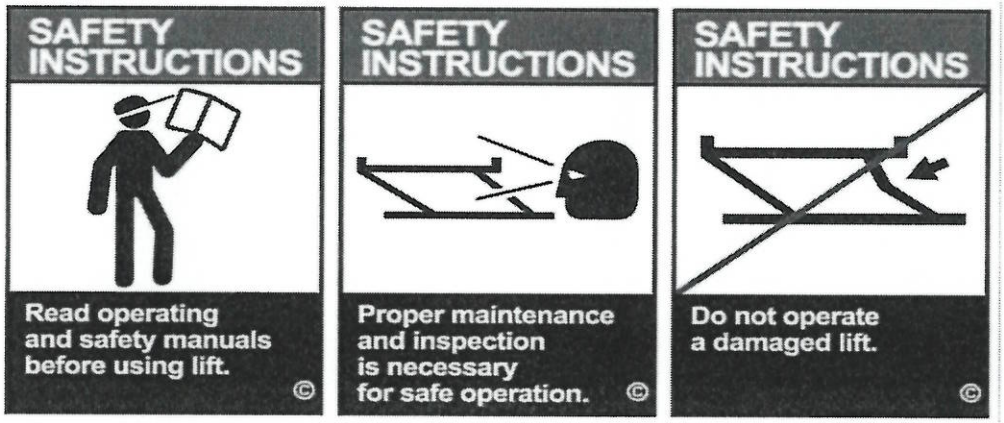
e) Do Not Exceed the Lift Capacity than rated.

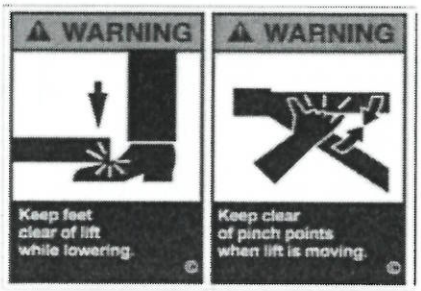
f) Before cleaning or servicing disconnect power supply

g) As the lift moves up and down, "pinch points" are formed. Keep hands, feet, hair, loose clothing, etc. away from these pinch points. Any of these may cause serious personal injury or death.

h) Ensure that the hydraulic, pneumatic and electrical lines will not be pinched by the lift as it rises or lowers. If a line gets pinched, the lift may not work properly. A hose may break, the lift table may drop suddenly, and someone may get injured. If an electrical line is broken, may cause personal injury or death due to electrical shock.

i) Read the motor nameplate and record the motor full load current. Consult local electrical codes for power supply fusing requirements. To avoid fire hazards, follow these fusing requirements.


SAFETY INSTRUCTIONS
Read operating and safety manuals before using lift.
SAFETY INSTRUCTIONS
Proper maintenance and inspection is necessary for safe operation.
SAFETY INSTRUCTIONS
Do not operate a damaged lift.


WARNING
Keep feet clear of lift while lowering.
WARNING
Keep clear of pinch points when lift is moving.


CAUTION
Lift to be used by trained operator only.
CAUTION
Authorized personnel only in lift area.

SECTION 8

8.DESCRIPTION .....Error! Bookmark not defined.

8.1 System Description.....Error! Bookmark not defined.

Fix the error

8.2 Operating Instructions .....Error! Bookmark not defined.

8. DESCRIPTION

8.1 System Description

- The hydraulic scissor lift load capacity rating and the serial number is mentioned on metal name plate.
- Lift should not be overloaded beyond the rated load capacity may result in damage.
- The hydraulic system consists of pump, motor, oil tank & valves.
- The gear pump is set at operating pressure of 200 bar.
- The operating principle is that the pump is operated to raise the table, and the pump is stopped when the table attains the desired elevation. A check valve and lowering valve between the pump and cylinder holds the table at elevation. For lowering to any desired level, the solenoid valve is energized to allow fluid to return from hydraulic cylinder to reservoir. A pressure compensated flow control valve is connected to the hydraulic cylinder to control the down speed at a predetermined rate under all load conditions.
- A scissor lift should only be operated on a solid surface such as concrete, tile, or a compacted road base. When lifts are placed on uneven surfaces or earth, mulch or gravel that is not compacted, the lifts may tip and fall over, causing serious injury or death.

Not sure where the number came about

1.1 MAIN SPECIFICATIONS

A) Maximum Load Lifting Capacity	9000 KG
B) Max Height of Platform	1960 MM
C) Actual Lift	1500 MM
D) Min Height of Platform	460 MM
E) Platform Length	3000 MM
F) Platform Width	2000 MM
G) Motor Capacity	10 HP
H) Oil Tank Capacity	100 LTR

8.2 Operating Instructions

- Check all electrical connection for short or phase short. Check hydraulic connections are tightened properly to ensure no oil leakage from joints.
- Connect (3 phase 415V) supply to the panel through cable top.
- Switch ON the RED LOTO SWITCH on panel & ensure Emergency push button is in unlock position.
- Use Lift's PUSH BUTTONS for moving lift upward & downward.
- Use Lip's PUSH BUTTONS for moving lip upward & downwards.
- In case of emergency, press emergency STOP push button provided on push button station.
- Before lifting Scissor, check motor rotates in clock wise direction.
- For controlling the lowering speed, adjust the valve provided on manifold.
- In case of failure & scissor lift platform is at top position, loosen the threads of manual override provided in the solenoid valve after reaching the platform at down position again retighten the same.

SECTION 9

9. EQUIPMENT SCHEDULES & ASSET REGISTERS 2

9.1 EQUIPMENT SCHEDULE 2

9.1.1 EQUIPMENT NAME 2

9.1.2 REFERENCE AND LOCATION 3

9.2 ASSET REGISTER 4

9. EQUIPMENT SCHEDULES & ASSET REGISTERS

9.1 Equipment Schedule

9.1.1 Equipment Name

Manufacturer	NIHVA TECHNOLOGIES PVT. LTD. PUNE, INDIA
A) Equipment Name	SCISSOR LIFT
B) Maximum Load Lifting capacity	9000 KG
C) Max. Height of Platform	1960 MM
D) Actual Lift	1500 MM
E) Min Height of Platform	460 MM
F) Platform Length	3000 MM
G) Platform Width	2000 MM
H) Motor Capacity	10 HP
I) Oil Tank Capacity	100 LTR

Confirm where the model number where taken from.

9.1.2 Reference and Location

S. No.	Reference	Model	System / Area Served	Location
1.	Drawing Ref. TDMZE- AES-AR-SDG-Z01-P01-1001	KHDL 710-20	Scissor Lift / Loading area	Ground floor – Gridline 1/ E to E' 2
2.				
3.				
4.				

9.2 *Asset Register*

Will be provided during final submission.

Asset Ref	Description	Category	Location				Make	Model	Serial No.	Quantity	Sub-Contractor	Supplier	Warranty Period
			Building	Level	Area	Room No.							
							KELLEYS	KHDL 710-20	ME12850	1	ASSA ABLOY Entrance Systems	NIHVA Technologies Pvt ltd	1 year

To be filled

Include power pack and control panel on the register.

SECTION 10

10. OPERATING INSTRUCTIONS AND PROCEDURES 2

10.1 START-UP PROCEDURE 2

10.2 NORMAL OPERATION 2

10.3 INSPECTION CHECK POINTS 2

10.4 HEALTH & SAFETY 2

10.5 TROUBLE SHOOTING GUIDELINES 2

10.6 SAFETY CHECK LIST 2

10. OPERATING INSTRUCTIONS AND PROCEDURES

10.1 Start-up procedure

- Check all electrical connection for short or phase short. Check hydraulic connections are tightened properly to ensure no oil leakage from joints.
- Connect (3 phase 415V) supply to the panel through cable top.
- Switch ON the RED LOTO SWITCH on panel & ensure Emergency push button is in unlock position.
- Use Lift's PUSH BUTTONS for moving lift upward & downward.
- Use Lip's PUSH BUTTONS for moving lip upward & downwards.
- In case of emergency, press emergency STOP push button provided on push button station.
- Before lifting Scissor, check motor rotates in clock wise direction.
- For controlling the lowering speed, adjust the valve provided on manifold.
- In case of failure & scissor lift platform is at top position, loosen the threads of manual override provided in the solenoid valve after reaching the platform at down position again retighten the same.

10.2 Normal operation

- Check all the electrical connections.
- Check the hydraulic connections.
- Ensure NO Oil Leakage from joints.
- Ensure the oil level on level indicator before using the lift.
- Before loading & unloading ensure the lip rests on truck properly.
- Ensure that there isn't any obstacle within the region of moving parts.
- While doing maintenance work, ensure that the Emergency switch is pressed & the maintenance stoppers are in their position while working below the lift.
- During brake down condition, use the LOTO switch.

10.3 Inspection Checkpoints

What is the frequency of inspection?

- Visually inspect all jack pins, pivot pins to ensure that the set screw has not loosened, allowing pins to move.
- Confirm Hex Nut Bolt, Allen Bolt -Pivot Pin Locking are Fixed firmly.
- Confirm Grub screw, Cup Washer on Scissor Arm Joint Bolt are fixed & are tightened firmly.
- Confirm Tightening of Rollers & its free movement.
- Confirm all Bolts at Scissor Arm are tightened.
- Visually inspect all lift arm rolling bearings to ensure that bearing bolts are tight.
- Visually inspect all hydraulic hoses and fittings to ensure there are no cut on hoses or leaking fittings.
- Visually inspect all electrical lines to ensure there are no cut wires or loose cords.
- Visually inspect the entire lift structure.
- These recommendations are applicable to the equipment considering daily usage of approx. 15-to 20 times.

- If usage is more frequent, the same time intervals for maintenance should be adjusted accordingly.

WEEKLY

- Do greasing / oiling of all moving joints.

MONTHLY

- Check hydraulic pipeline for leak proof working.
- Check oil level in hydraulic tank.
- Check cylinder seals & 'O` rings for leak proof working.

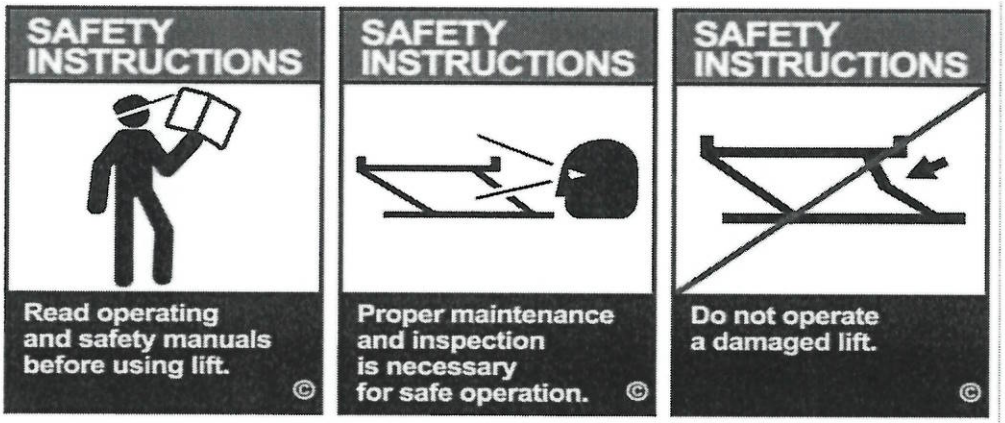
YEARLY

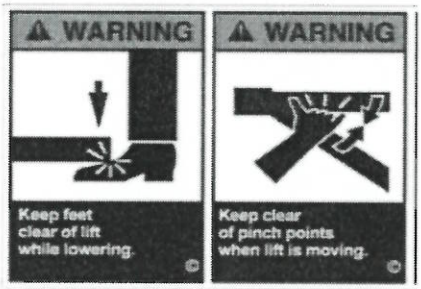
- Change hydraulic oil.
- Check main shaft seal & shaft 'O` ring for leak proof working.
- Hydraulic oil to be used – Enclo – 68 OR Servo – 68.

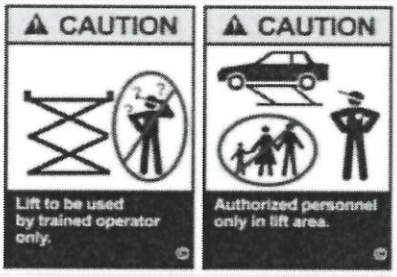
10.4 Health & Safety

Use trained personnel for loading & unloading of equipment. The equipment must be properly handled when transferring from the unloading area to final site to prevent damage. DO NOT operate equipment unless it is properly anchored. It may cause serious injury or death.

- Install all safety devices, guards and guarding prior to equipment start-up.
- Only Trained & Authorized person must use the lift.
- Always use mechanical stopper if you are going to work in elevated condition.
- Keep working area free from oil to Avoid Slipping/Fire hazards.
- Do Not Exceed the Lift Capacity than rated.
- Before cleaning or servicing disconnect power supply
- As the lift moves up and down, "pinch points" are formed. Keep hands, feet, hair, loose clothing, etc. away from these pinch points. Any of these may cause serious personal injury or death.
- Ensure that the hydraulic, pneumatic and electrical lines will not be pinched by the lift as it rises or lowers. If a line gets pinched, the lift may not work properly. A hose may break, the lift table may drop suddenly, and someone may get injured. If an electrical line is broken, may cause personal injury or death due to electrical shock.
- Read the motor nameplate and record the motor full load current. Consult local electrical codes for power supply fusing requirements. To avoid fire hazards, follow these fusing requirements.


SAFETY INSTRUCTIONS
Read operating and safety manuals before using lift.
SAFETY INSTRUCTIONS
Proper maintenance and inspection is necessary for safe operation.
SAFETY INSTRUCTIONS
Do not operate a damaged lift.


WARNING
Keep feet clear of lift while lowering.
WARNING
Keep clear of pinch points when lift is moving.


CAUTION
Lift to be used by trained operator only.
CAUTION
Authorized personnel only in lift area.

10.5 Trouble shooting Guidelines

i. If lift will not raise:

- Check electric circuit on new installations of 3-phase motors, if lift does not start raising in approximately 10 seconds, reverse two power supply leads to reverse pump rotation. (if motor direction is reverse)
- Check oil level. With lift in full up position, oil level should be
- Approximately 1¼" above tank bottom.

ii. *Lift raises slower than specified rate or will raise only partial load:*

Note: Allowance must be maintained for slight speed variations (due to temperature which changes the viscosity of the hydraulic oil varies)

- Check line voltage under load condition. Low voltage affects capacity of equipment.
- If foam is visible in tank oil, check for a loose connection in suction line between pump and tank. Tighten connections.

iii. Lift slowly settles with lowering valve not energized:

- Flush lowering valve by cycling lift up and down. This should be done with no load.
- If lift continues to settle, replace down valve cartridge.
- If lift continues to settle, replace pump.

iv. If Lift will not lower:

- Look for mechanical restrictions.
- Test valve coil for operation. Check voltage at coil.
- If voltage is ok, replace coil.

v. If lift will not raise to full vertical travel:

a. Check for low oil level if necessary, add oil.

vi. If lift does not lower smoothly:

a. Run the lift up and down under load condition for some time to purge air.

10.6 Safety Check List

- Leakage at Piping –
  - Tighten Joints.
  - Replace Dotty Seal at Mounting

- Leakage in Cylinder-
  - Replace Seals.

- Replace Oil after Every Year-
  - Drain the Oil from the System.
  - Clean filter, Tank elements, and strainers before refilling.

SECTION 11

11. METHOD STATEMENT FOR SCISSORLIFT 2

Not related to section "Operating manual and literature"

11. Method statement for scissor lift

- Check all pit dimensions for accuracy.
- Attach a temporary electrical line through the pit conduit to the lift. Check for correct motor rotation
- Using slings, encircle the entire lift, not just the platform and lower the lift into the pit centering it for 1" minimum clearance on all sides to the pit wall.
- Raise the lift with the push button and remove the slings. Run the unit up and down several times to remove air from the hydraulic system.
- Level and center the lift by shimming and grouting the entire base frame, not just the corners. Lag the unit in place using M16 x 150L wedge anchors in the holes provided. Caution! Continuous base frame support is essential for proper installation
- With the lift fully elevated, disconnect the main power and complete the permanent electrical wiring.
- Follow the instructions outlined in operating manual to complete the installation.

SECTION 12

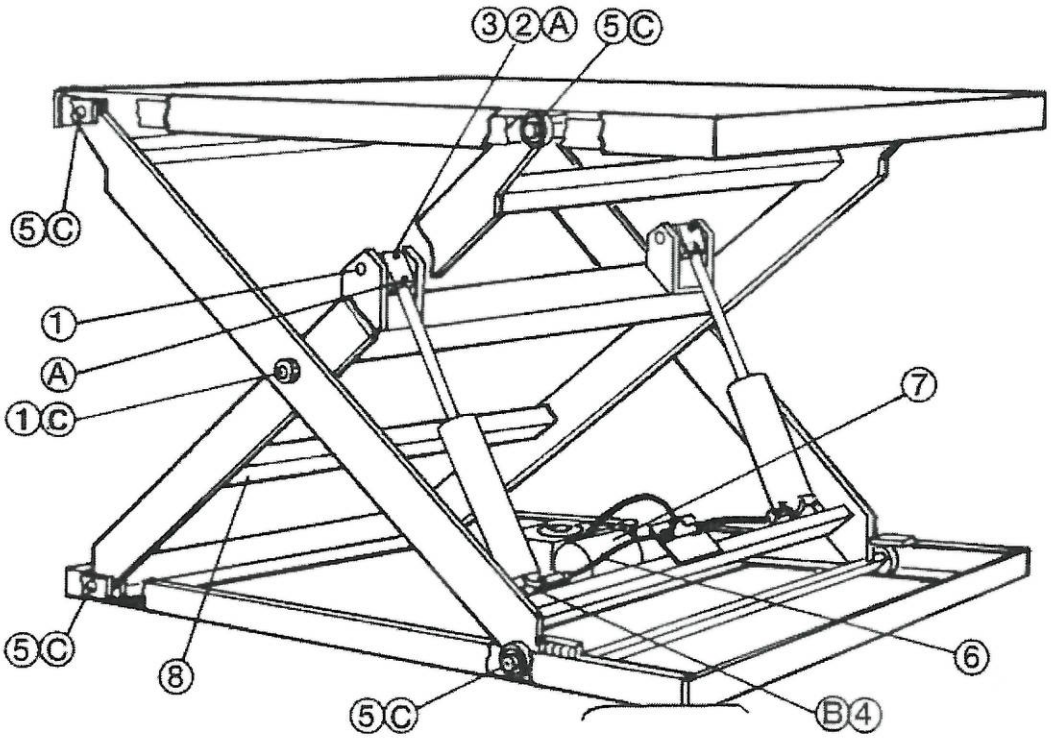
12. MAINTENANCE PROCEDURES 2

12.1 PLANNED MAINTENANCE PROCEDURES 2

12.2 PLANNED MAINTENANCE SCHEDULES 2

12. MAINTENANCE PROCEDURES

12.1 Planned Maintenance Procedures



Lubrication Points

Lubrication points as mentioned below,

- Points noted with 'A', apply grease lubrication.
- Points noted with 'B', apply oil lubrication.
- Points noted with 'C', no lubrication required (Pre-Lubricated &sealed)
- Ensure all Moving parts like Wheel, Flap Joint, Rollers etc. are greased.

Maintenance procedure for the power pack and control panel are to be included.

12.2 Planned Maintenance Schedules

- **WEEKLY**
- Do greasing / oiling of all moving joints.

- **MONTHLY**
- Check hydraulic pipeline for leak proof working.
- Check oil level in hydraulic tank.
- Check cylinder seals & 'O` rings for leak proof working.

- **YEARLY**
- Change hydraulic oil.
- Check main shaft seal & shaft 'O` ring for leak proof working.

- Hydraulic oil to be used – Enclo – 68 OR Servo – 68.

SECTION 13

13. SPARE AND REPLACEMENT PARTS 2

13. SPARE AND REPLACEMENT PARTS

S. No.	Recommended	Actual Installed	System	Part Description And Model Number	Location / Area	Units Issued	Supplier	Delivery period to site from order	Current Unit Price (AED)	Shelf Life or special storage conditions
1			Electro Mechanical	Scissor Lift	Ground floor grid 1-E to E'	1	ASSA ABLOY Entrance Systems			Not applicable
		Include power pack and control panel.								

SECTION 14

14. DISPOSAL INSTRUCTIONS 2

14.1 MAINTENANCE DISPOSAL / DEMOLITION 2

14. DISPOSAL INSTRUCTIONS

14.1 Maintenance Disposal / Demolition

SECTION 15

15. TESTING & COMMISSIONING DATA 2

15. TESTING & COMMISSIONING DATA

The final Testing and Commissioning Data is attached.

Attachment should be along with the section.

SECTION 16

16. MODIFICATIONS 2

16.1 MANUFACTURERS MODIFICATION DETAILS 2

16. MODIFICATIONS

16.1 Manufacturers Modification Details

NA

16.2 Designers Modification Details

NA

SECTION 17

17. AS BUILT DRAWINGS 2

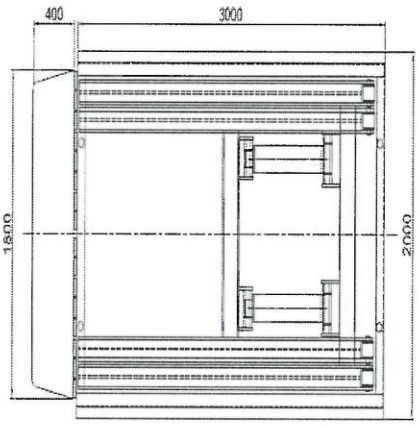
Include all related drawings references.

17. AS BUILT DRAWINGS

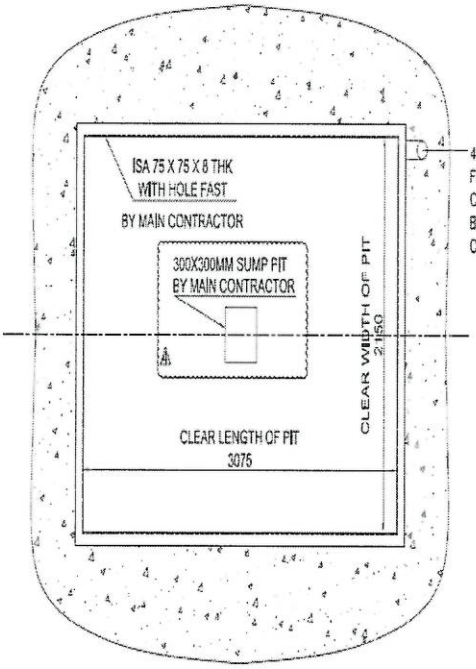
The final As-built drawings will be included in the Final Submission of this O&M Manual.

Drawing no. is not as per the project protocol.

Drawing No.	Drawing Title / Description
TDMZE-AES-AR-SDG-Z04-P01-1001	Electro Hydraulic Scissor Lift
NIHVA/18-19/VDR/291	POWER PACK
NIHVA/18-19/VDR/292	HYDRAULIC CIRCUIT
NIHVA/18-19/VDR/288	ELECTRICAL CIRCUIT

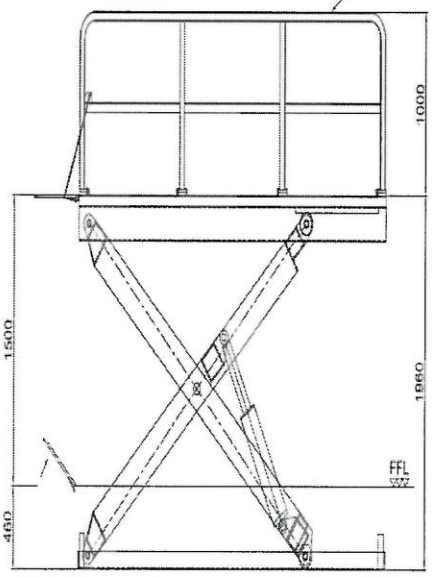


PLAN VIEW

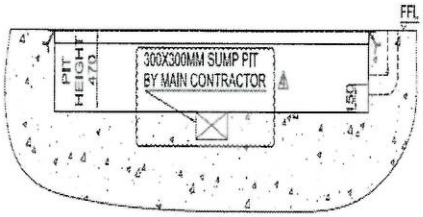


TOP VIEW OF PIT

SAFETY RAILING
ON TWO LENGTH SIDES

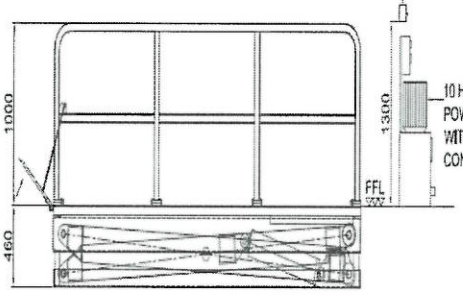


FRONT VIEW - OPEN HEIGHT



FRONT VIEW OF PIT

360V/3PH/50Hz
ISOLATOR BY MAIN
CONTRACTOR

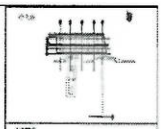


FRONT VIEW - SHUT HEIGHT

NOTE : PIT LEVEL SHOULD PARALLEL.
GENERAL TOLERANCE - ±10mm
PIT DIAGONAL SHOULD BE ±3mm

FOR WELDING DETAILS REFER TO
DRAWING No. TDMZE-AES-AR-SDG-Z01-P01-1003

CAPACITY	9 TON
PLATFORM SIZE	3000(L) X 2000(W)
POWER PACK	10 HP, 3 PH
SHUT HEIGHT	460 MM
LIFT HEIGHT	1500 MM
OPEN HEIGHT	1960 MM



NOTES

THE DUBAI MALL - ZA'ABEEL EXPANSION

EMAAR PROPERTIES P.J.S.C.

Turner

Dutco Construction Co. LLC

ASSA ABLOY Entrance System LLC

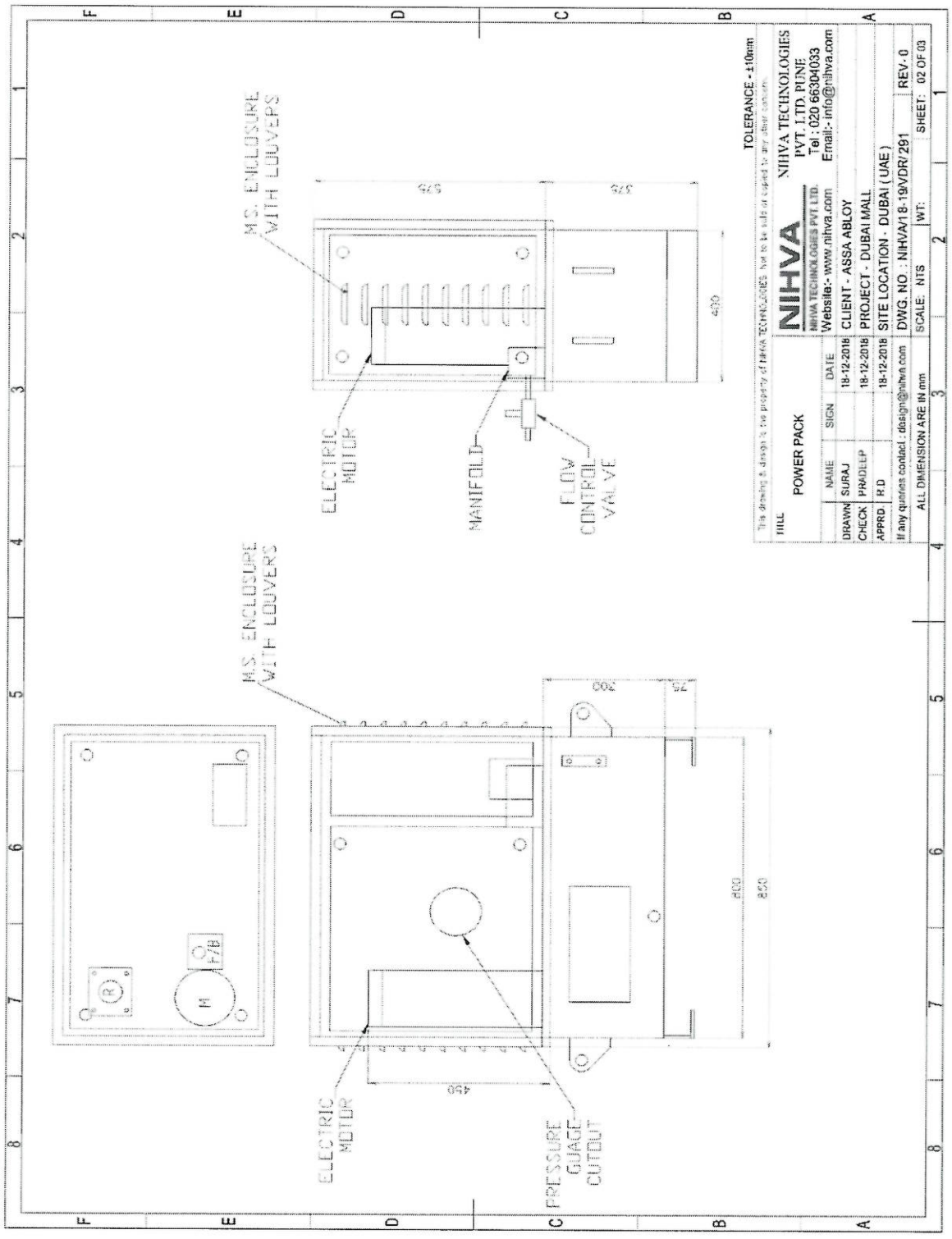
DUBAI
P.O. BOX: 62553
TEL: 04-8862288
FAX: 04-8864419

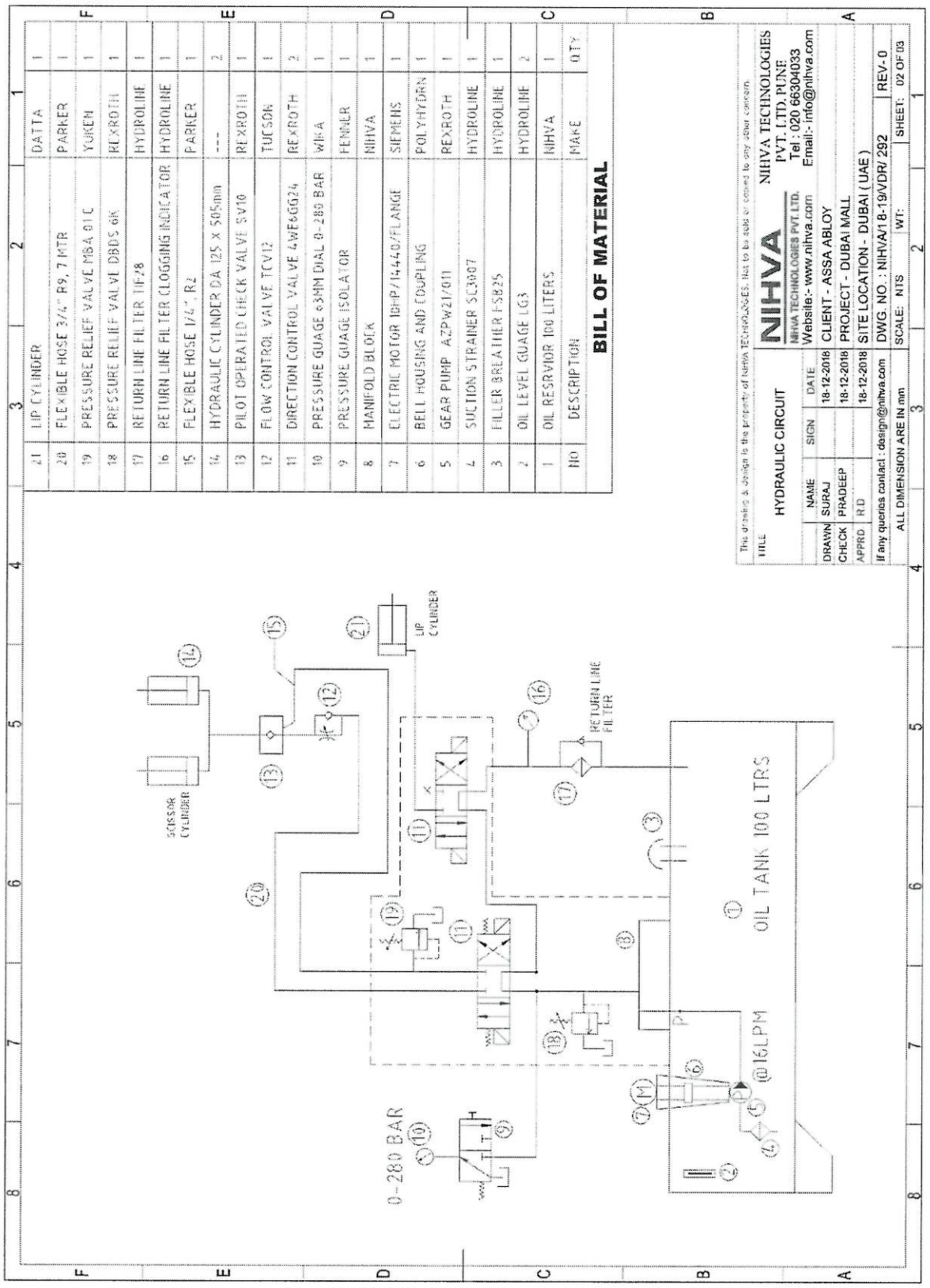
ABU DHABI
P.O. BOX: 32286
TEL: 02-4414033
FAX: 02-4413937

ELECTRO HYDRAULIC SCISSOR LIFT

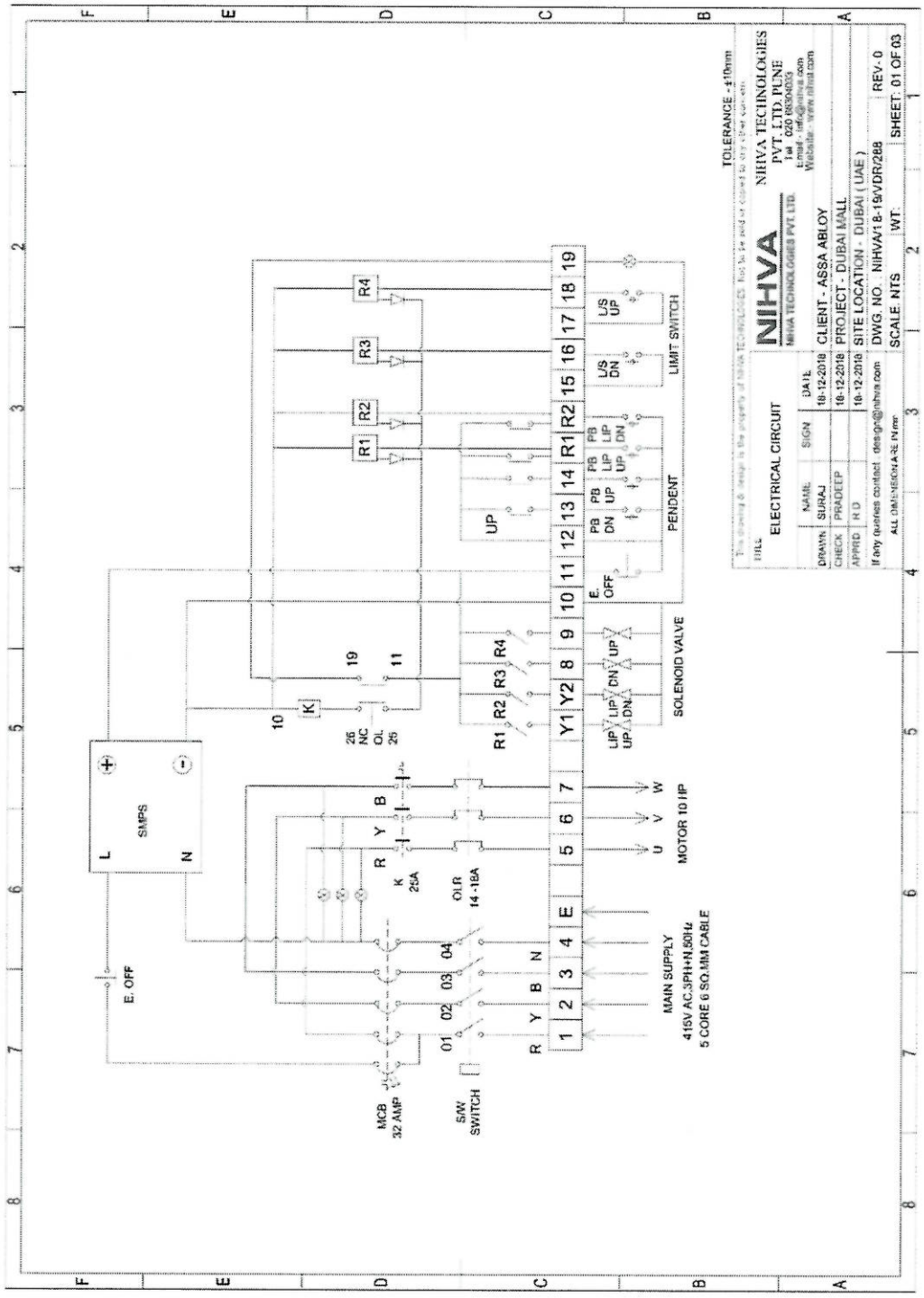
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1. HYDRAULIC POWER PACK & CIRCUIT: -





2. ELECTRICAL CIRCUIT:-


NIHVA
NIHVA TECHNOLOGIES PVT. LTD. PUNE
ELECTRICAL CIRCUIT
CLIENT - ASSA ABLOY
PROJECT - DUBAI MALL
SITE LOCATION - DUBAI (UAE)
DWG. NO. - NIHVA/18-19/VDR/288
SCALE: NTS
SHEET: 01 OF 03
REV- 0
SMPS
MCB 32 AMP
S/W SWITCH
E. OFF
MAIN SUPPLY 415V AC,3PH+N,50Hz 5 CORE 6 SQ.MM CABLE
MOTOR 10 HP
SOLENOID VALVE
PENDENT
LIMIT SWITCH

SECTION 18

18. MANUFACTURER AND SUPPLIER LITERATURE 2

18. MANUFACTURER AND SUPPLIER LITERATURE

Manufacturers' Literature Index		
Manufacturer / Equipment	Supplier	Contact Details
NIHVA TECHNOLOGIES PRIVATE LIMITED, PUNE 411019, INDIA	**Supplier:** ASSA ABLOY Entrance Systems UAE P O Box 80983 Dubai United Arab Emirates	Tel: +971 4 811 8100 Fax: +971 4 886 5465 E-mail: philipose.varughese@emerson.com Web: www.mvptech.ae
	Manufacturer : NIHVA TECHNOLOGIES PRIVATE LIMITED, PUNE, INDIA G-BLOCK, 1ST FLOOR, THERMAX CHOWK MIDC, ABOVE RENAULT SHOWROOM CHINCHWAD, PUNE-411019, INDIA	Tel: **+912066304033** E-mail: info@nihva.com Web: www.nihva.com

Publication Details	No. of Pages
User Manual for Scissor Lift	

SECTION 19

19. PRODUCT GUARANTEES & WARRANTIES 2

19.1 WARRANTEE LETTERS 2

19. PRODUCT GUARANTEES & WARRANTIES

19.1 Warrantee Letters

All Warranty Letters will be Included in the Final Submission of this O&M Manual.

SECTION 20

20. CROSS REFERENCING INDEX.....2

20. CROSS REFERENCING INDEX

Our Project Reference number is ME12850

Not related on the above section.